For the second half of a cycle, from 15 ms to 25 ms, the diode is reverse biased and the p.d. across the load resistor is zero.

8 **(a)** $T = 4.70 - 1.20 = 3.50$ days

(b) **(i)** Gravitational force between both stars provides the centripetal force required for the circular motion. *(this statement is sufficient to get 1 mark for the explanation part)*

$$\frac{GM_SM_P}{d^2} = M_P y\omega^2$$

$$M_S = d^2 y\left(\frac{2\pi}{T}\right)^2\left(\frac{1}{G}\right)$$

$$M_S = \left(\frac{4\pi^2}{G}\right)\frac{yd^2}{T^2} \quad \text{(shown)}$$

$$\frac{GM_SM_P}{d^2} = M_S x\omega^2$$

$$M_P = d^2 x\left(\frac{2\pi}{T}\right)^2\left(\frac{1}{G}\right)$$

$$M_P = \left(\frac{4\pi^2}{G}\right)\frac{xd^2}{T^2} \quad \text{(shown)}$$

(ii)

$$M_S + M_P = \left(\frac{4\pi^2}{G}\right)\frac{yd^2}{T^2} + \left(\frac{4\pi^2}{G}\right)\frac{xd^2}{T^2}$$

$$= \left(\frac{4\pi^2}{G}\right)\frac{d^2}{T^2}(y+x)$$

Since $d = x + y$

$$M_S + M_P = \left(\frac{4\pi^2}{G}\right)\frac{d^3}{T^2} \quad \text{(shown)}$$

(c)

$$M_S + M_P = \left(\frac{4\pi^2}{G}\right)\frac{d^3}{T^2}$$

$$2.28\times10^{30} = \left(\frac{4\pi^2}{6.67\times10^{-11}}\right)\frac{d^3}{(3.50\times24\times3600)^2}$$

$$d = 7.06\times10^9 \text{ m}$$

(d)

$$v_P = \frac{2\pi r}{T}$$

$$= \frac{2\pi\times7.062\times10^9}{3.50\times24\times3600}$$

$$= 1.467\times10^5 = 1.47\times10^5 \text{ m s}^{-1}$$

(e) Since the center of mass of the system is stationary, the momentum of the system must be zero.

$2.28 \times 10^{30} \times 84.3 + M_P \times (-1.467 \times 10^5) = 0$

$M_P = 1.31 \times 10^{27}$ kg

(f)
1. The plane of orbit of the planet must be such that the transit can be observed from Earth.
2. The size of the planet must be large enough to block enough starlight.
3. The time of transit across the star must be long enough to be detectable.
4. Exoplanets with very long periods may not be detected as the chances of a transit happening while the stars are under observation is very small or may not happen in our lifetime.
5. The brightness of the star must be large enough so that a change in the brightness can be detected.
6. Rogue planets cannot be detected as they do not orbit around any stars.

(g) One light-year = speed of light × one year = $3.0 \times 10^8 \times 365 \times 24 \times 3600$
$= 9.46 \times 10^{15} = 9.5 \times 10^{15}$ m

(h) **(i)** Rayleigh criterion states that <u>two images are just resolved</u> when the <u>central maximum of **the diffraction pattern** of one image falls on the first minimum of the **diffraction pattern of the other image**</u>.

(ii)

$$S = r\theta$$

$$7.06 \times 10 = \left(9.46 \times 10^{15} \times 159\right) \times \theta$$

$$\theta = 4.69 \times 10^{-9} \text{ rad}$$

OR

$$\tan\frac{\theta}{2} = \frac{7.06 \times 10^9 / 2}{9.46 \times 10^{15} \times 159}$$

$$\theta = 4.69 \times 10^{-9} \text{ rad}$$

(iii) Using Rayleigh's criterion,

$$\theta_{\text{min}} = \frac{\lambda}{b} = \left(\frac{14 \times 10^{-6}}{6.5}\right)$$

$$= 2.154 \times 10^{-6} \text{ rad}$$

Since <u>θ is smaller than θ_{min}</u>, the telescope will <u>not</u> be able to distinguish Osiris from the star.